

**THE ANTHELMINTIC EFFECT OF *ARTEMISIA VESTITA* ON
HAEMONCHUS CONTORTUS- AN ABOMASAL
BLOODSUCKING NEMATODE OF SMALL RUMINANTS**

DISSERTATION

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Submitted By

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DEPARTMENT OF ZOOLOGY
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UNIVERSITY OF KASHMIR,
HAZRATBAL, SRINAGAR-190006
(NAAC Accredited Grade 'A++')

2025



DEPARTMENT OF ZOOLOGY
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CERTIFICATE

This is to certify that the project work entitled “**The anthelmintic effect of *Artemisia vestita* on *Haemonchus contortus*- an abomasal bloodsucking nematode of small ruminants**” is based on the work carried out by **Ms. Maria, Enrollment No. 23061119001**, batch 2023 in the 4th semester under course MSc Zoology under my guidance and supervision from March 2025 to September 2025 at Department of Zoology, School of Biological Sciences, University of Kashmir, Srinagar- 190006, J&K.

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Declaration by the Candidate

I hereby declare that the dissertation entitled “**The anthelmintic effect of *Artemisia vestita* on *Haemonchus contortus*- an abomasal bloodsucking nematode of small ruminants**” submitted to Department of Zoology, University of Kashmir is my original work carried out under the supervision of **Prof. Fayaz Ahmad**, Professor Department of Zoology and Director of Admissions and Competitive Examinations.

I further declare that this work has not been submitted earlier to any other University or Institution for the award of any degree or diploma. I also declare that all sources of information and references used in this dissertation have been properly acknowledged and cited.

Maria

Dated.....

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ABSTRACT

Haemonchus contortus commonly known as barber's pole worm is one of the most pathogenic gastrointestinal nematodes of small ruminants causing severe anemia, reduced productivity and significant economic loss globally. It primarily infects the abomasum of small ruminants such as sheep and goats. It is considered one of the most economically important parasites in tropical and subtropical regions due to its blood feeding habit. The life cycle of *H. contortus* involves several stages from egg to larva and ultimately to the adult worm. The animals get infected by grazing on the pastures contaminated with infected larvae. After entering the body, the worms established themselves and start to feed on the blood leading to anemia, weight loss, decrease in milk production and in severe cases fatalities particularly in vulnerable population.

The control of *H. contortus* has traditionally relied on synthetic anthelmintics, however widespread and repeated use of these drugs has led to the development of multi drug resistance making management increasingly difficult. This challenge has prompted growing interest in alternative strategies, and this is necessitating a shift towards integrated parasite management strategies. Emergence of resistant strains as well as high cost of currently available drugs has led to the evaluation of other alternative helminth control options mainly from medicinal plants.

Artemisia vestita, a medicinally important plant species growing in Kashmir is used traditionally by people as vermifuge in addition to its other livestock uses. The current study was designed to evaluate invitro anthelmintic activity of *A. vestita* against *H. contortus* in comparison with albendazole to investigate the effect of plant extracts on the survival of adult worm. Phytochemical screening of extracts was conducted to confirm the presence of bioactive compounds like alkaloids, flavonoids, phenolic compounds etc. In this study aqueous and methanolic extracts of *A. vestita* were used at various concentrations. Methanolic extract showed more promising effect on *H. contortus* mortality as compared to aqueous extract. The observed efficacy is attributed to the synergistic activity or bioactive compounds which may interfere with nematode metabolism and cuticular integrity. These findings indicate that *A. vestita* possess promising anthelmintic properties against *H. contortus* and could serve as a basis for the development of plant based alternatives to synthetic drugs. These findings could be further used as a future strategy for management of this parasite.

Key words: *Artemisia vestita*, anthelmintic activity, bioactive compounds, *Haemonchus contortus*, phytochemical screening.

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1. INTRODUCTION

Haemonchus contortus is an abomasal blood sucking nematode which is prevalent in sheep goat and many wild ruminants in most parts of the world. It is a highly pathogenic nematode parasite causing severe anemia in heavy infections. Due to its blood feeding behavior, it is a common cause of mortalities in goats, sheep and occasionally other ruminants (Besier *et al.*, 2016). The parasite is present throughout the tropical and subtropical regions of the world where it is a major constraint for profitable production of sheep and goats (Zenebe *et al.*, 2017). The infection caused by *H. contortus* is called haemonchosis. The severity of the disease depends on a variety of factors including the number of helminths infecting an animal (intensity of the infection), isolated-related parasite virulence, host age, animal breed and the nutritional and immunological status of infected sheep (Francisco *et al.*, 2007).

H. contortus is one of the most fecund strongyle nematodes where individual females can produce thousands of eggs per day that can lead to rapid larval-pasture contamination and associated outbreaks of haemonchosis (Jesse *et al.*, 2019). Sheep are usually more prone to gastrointestinal tract parasitism as they mostly graze on the pastures contaminated with L3 of parasite nematodes (Tariq *et al.*, 2008). In the life cycle of *H. contortus* egg and larval stages are found outside their host, whereas adults stages live inside their host (Fig. 1). During times of high humidity and high temperatures, greater number of L3 tend to be found higher upon vegetation (Silva *et al.*, 2008). The young cattle are more prone to L3 infections and the primary consequences of *H. contortus* infection include anemia, leading to significant blood loss in infected animals (Flay *et al.*, 2002). One characteristic sign of *H. contortus* infection is the development of bottle jaw or submandibular edema. This swelling under the jaw is a result of fluid accumulation and is a common clinical sign of advanced haemonchosis (Fávero *et al.*, 2016). The bottle jaw is caused by a chronic shortage of protein in the bloodstream of host. *H. contortus* infections are commonly identified as the most significant (Emery *et al.*, 2016; Jamalm *et al.*, 2016) with significant rates of growth and milk yield reduction in small ruminants (Andrea *et al.*, 2011). The stress caused by the parasitic infections can impact reproductive performance, resulting in decreased conception rates and lamb or kid survival (Saccareau *et al.*, 2017). Both body weight and reproductive status of the host influence the parasitic infection development due to the development of acquired

immunity and gradual increase in weight along with the age of animals (Tasawar *et al.*, 2010).

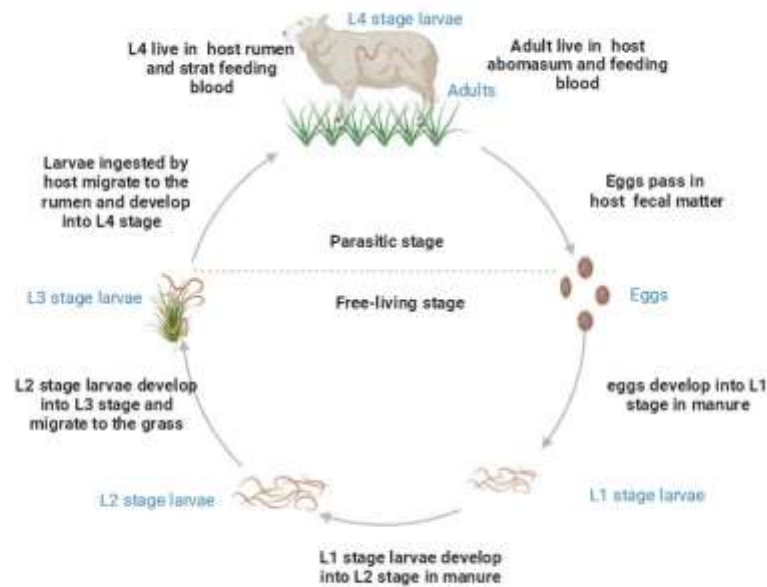


Fig. 1. Several stages of life cycle of *Haemonchus contortus*

The high pathogenicity of *H. contortus* is attributed to its high fecundity, immune evasion mechanisms and its ability to develop resistance against routinely administered anthelmintic drugs. The strategy for the control of *H. contortus* has traditionally been the administration of anthelmintic compounds. The most commonly used synthetic drugs against *H. contortus* include benzimidazoles e.g., albendazole, fenbendazole and thiabendazole (Stasiuk *et al.*, 2019). It should be emphasized that the excessive and uncontrolled use of anthelmintic drugs may result in the emergence of resistance in *H. contortus*. In addition, the high costs, limited availability of these chemicals and the drug residues in the final products and the environment following their use or the other factors that discourage many farmers from using them in some emerging countries (Knox *et al.*, 2006). According to the pharmaceutical companies the annual cost of antiparasitic compounds is proposed to be tens of billions of dollars worldwide (Wolstenholme *et al.*, 2004). Macrocyclic lactones and benzimidazoles are the most frequently used anthelmintics in the sheep industry and the resistance against them has been reported worldwide. Target site related mechanisms usually are investigated first. For benzimidazoles mutations in β -tubulin isotype -1 gene leading to the amino acid changes

at the drug target site have been found to confer resistance in trichostrongyloid nematodes (Yilmaz, 2019). Macrocyclic lactones enhance the release of gamma-aminobutyric acid (GABA), an inhibitory neurotransmitter leading to the paralysis and death of the parasite. They also effect the worm's reproductive system (Menez *et al.*, 2012). Drug efflux by p-glycoproteins is a non-target site related mechanism that has been proposed to play a role in macrocyclic lactone resistance particularly (Yilmaz, 2019). The overuse of these drugs and their similar modes of action can contribute to cross resistance between different macrocyclic lactones (Reyes- Guerrero *et al.*, 2020). The anthelmintic drug levamisole is thought to bind to nicotinic acetylcholine receptors of nematodes. It is possible that resistance to this drug is associated with either a change in binding characteristics or a reduction in the number of nicotinic acetylcholine receptors (Hoekstra *et al.*, 1997). Newer drugs like monepantel have been introduced to combat resistance, however cases of reduced efficacy have been reported, indicating the potential for resistance development (Kotze *et al.*, 2016).

A dynamic and innovative approach to managing *H. contortus* is critical to reducing dependence on veterinary anthelmintics (Kearney *et al.*, 2016). The drug resistant strains of parasites have necessitated the development and the use of various parasitic control methods such as grazing management, biological agents and vaccines, selection of resistant breeds of animals with or without moderate use of anthelmintics (Getachew *et al.*, 2007). Selective breeding for resistance traits in animals can be effective long-term strategy (Gowane *et al.*, 2020).

Anthelmintics derived from plants can be an alternative for the treatment of parasitic infections (Akhter *et al.*, 2000). The use of medicinal plants for the prevention and treatment of gastrointestinal parasitism has its origin in ethnoveterinary medicine (Athanasiadou *et al.*, 2007). Plant based bioactive compounds known as phytochemicals are considered useful alternatives or supplements for controlling *H. contortus* in livestock. Many of these natural compounds show anthelmintic effects and can be applied within integrated parasite management programs. Studies from various parts of the world have shown that certain plant species effectively reduce the degree of parasite infestation in sheep and are promising alternatives to conventional anthelmintics (Githiori *et al.*, 2006). There is a dire need to develop alternative anthelmintic approaches from natural flora which can be less toxic, biodegradable, environment friendly and to

cover the most challenging problem of parasite resistance issue (Carvalho *et al.*, 2012). Worldwide different plant species have been reported and evaluated for natural bioproducts to control the parasitic infections and reduce the dependency on conventional chemotherapy (Ali *et al.*, 2021). The identified plant species with promising *invitro* anthelmintic activity against *H. contortus* are *Lachesiodendron viridiflorum*, *Cymbopogon schoenanthus*, *Allium sativum*, *Artemisia herba-alba*, *Corymbia citriodora*, *Artemesia absinthium* etc. The antiparasitic activity could be attributed to the presence of potent phytochemicals such as saponins, tannins, flavonoids in Asteraceae (Carvalho *et al.*, 2013), saponins, essential oils in Lamiaceae (Raja, 2012), phenolic compounds, alkaloids, hemagglutinins in Fabaceae (Zarnowski *et al.*, 2001) and alkaloids and terpenoids in Euphorbiaceae (Mwine and Van Damme, 2011). Mechanism of action of these compounds against different developmental stages of *H. contortus* is through the inhibition of egg embryonation and direct effects on the larva (Al-Shaibani *et al.*, 2008). The antiparasitic activity of condensed tannins is attributed to their ability to bind with the cellular membrane proteins, which result in unavailability of nutrients to the larva causing starvation and death (Athanasiadou *et al.*, 2001).

Asteraceae is the first and Fabaceae is the third largest terrestrial plant families all over the world and this could be the possible reason for such an extensive utilization of these families (Tariq *et al.*, 2017). *Artemisia absinthium* belonging to the family Asteraceae contains a compound artemisinin that can inhibit vital enzymes of metabolic cascade by forming the covalent bonds and resulting in irreversible inhibition of enzyme activities (Wang *et al.*, 2015). Artemisinin has also been reported in other species of *Artemisia* including *Artemisia vestita*. Artemisinin causes the disruption of mitochondrial membrane potential causing cytochrome-c release into the cytoplasm (Jia *et al.*, 2016) and inhibit the electron transfer and oxidative phosphorylation of mitochondria, with final activation of caspase-3 mediated apoptosis (Li *et al.*, 2005).

The crude aqueous extract of *Artemisia* species can induce ultrastructural changes such as tegument damage, lipid accumulation, glycogen depletion and finally worm paralysis and death (Beshay, 2018). Chemical analysis of *Artemisia absinthium* has shown that its volatile oil is rich in thujone (α and β) which has been earlier reported as an anthelmintic (Meschler and Howlett, 1999). Phytochemical studies of *Artemisia vestita* has shown the presence of camphor, eucalyptol and thujone which are predominantly present in other

species of *Artemisia* (Dogra *et al.*, 2024). It is essential to emphasize that although phytochemicals demonstrate potential their effectiveness can differ, and further investigation is required to establish appropriate dosages and formulations for effective use. Furthermore, incorporating phytochemicals into a comprehensive parasite management framework is necessary for sustainable and long-term control.

Within this context the current study was carried out with the aim to evaluate the *invitro* anthelmintic activity of *A. vestita* in comparison to albendazole against the *H. contortus*. The *invitro* anthelmintic activities of crude aqueous and methanolic extracts of *A. vestita* were investigated on adults of *H. contortus*. Phytochemical analysis revealed that overall profile of both extracts was dominated by flavonoids, tannins and terpenoids. Other secondary metabolites include alkaloids, glycosides, phenolic compounds and quinones. The methanolic extract showed better *invitro* activity against adult parasites than the aqueous extract in terms of paralysis and death of worms at different hours post treatment. The different concentrations of solvent extracts were compared for their efficacy. The comparative evaluation revealed a clear concentration dependent anthelmintic activity. The current study was undertaken to provide the experimental evidence to support its potential use as a natural alternative in the control of *H. contortus*.

2. REVIEW OF LITERATURE

Nematodes of genus *Haemonchus* are cosmopolitan parasites of abomasum of cattle, sheep, goats and a variety of ruminants and clinical disease caused by large number of these helminths is a well-known syndrome manifested by severe anemia, hypoproteinemia, weakness and death (McGhee *et al.*, 1981). The control of *H. contortus* often involves the use of anthelmintic drugs however it is important to note that indiscriminate and frequent use of anthelmintics can lead to development of the resistance. There is evidence that the incidence of anthelmintic resistance is increasing in livestock in countries throughout the world (Jackson, 1993). Anthelmintic resistance to the benzimidazole has been reported in *H. contortus* in Sweden (Nillson *et al.*, 1993). A study in this country has shown that *H. contortus* has undergone some unique epidemiological adaptations (Waller *et al.*, 2004). Thus, the need for the sustainable control strategies for *H. contortus* is becoming much more pressing (Waller and Chandrawati, 2005).

The development of anthelmintic resistance led to the evaluation of medicinal plants as the alternative source of anthelmintics (Egualé *et al.*, 2009). Ethnoveterinary surveys were conducted to compile a comprehensive inventory of plants and employed in the treatment of nematodiasis. 85 Species of Australian native shrub extracts were measured for invitro anthelmintic activity and approximately 40% of the species showed significant activity in inhibiting development of *H. contortus* larvae. Plant reproductive maturity has been associated with the increasing anthelmintic activity in some species (Kotze *et al.*, 2009). Costa *et al.* (2008) has evaluated extracts of *Azadiractita indica* (neem) for their anthelmintic potential against *H. contortus*. *Invitro* assays revealed that both ethyl acetate and ethanol extracts inhibited egg hatching and larval development in a dose dependent manner with the ethanol extract showing strongest activity. Hernandez-Villegas *et al.* (2011) investigated different solvent extracts of *Phytolacca icosandra* against *H. contortus*. At concentration as low as 0.15mg/ml the ethanol extract suppressed more than 70% of egg hatching while the dichloromethanol fraction exhibited an LD50 of 0.28mg/ml achieving over 90% inhibition at higher doses. Medicinal plants frequently evaluated for their potential against small ruminant parasites include *Terminalia leiocarpa*, *Momordia charantia*, *Caesalpinia bonduc*, *Alchornea febrifuga*, *Jatropha multifida*, *Vitex doniana* and *Zanthoxylum zanthoxyloides*. Several plants investigated in

this study has shown significant effectiveness and have been widely examined for their anthelmintic potential. A notable example is *Z. zanthoxyloides* which has been rigorously studied both *invitro* and *invivo* for its ability to control gastrointestinal parasites of small ruminants including *H. contortus* and *Trichostrongylus colubriformis* (Azando *et al.*, 2011). Chakraborty *et al.* (2014) demonstrated anthelmintic properties of *M. charantia* against many gastrointestinal nematode parasites.

Different *Artemisia species* have also been investigated for their anthelmintic potential against *H. contortus*. In a comparative study with ivermectin *Artemisia vestita* and *Artemisia maritima* extracts significantly reduced larval and adult survival produced notable fecal egg count (FEC) reduction in infected sheep (Irum *et al.*, 2015). A controlled clinical trial has been carried out by Da-silva *et al.* (2017) to evaluate the effect of oral administration of *Artemisia annua* in sheep experimentally infected with *H. contortus*. It exhibited an average efficacy of 89% as an anthelmintic agent in sheep. The anthelmintic property of the medicine plants including the *Artemisia* is attributed to the presence of phytochemicals or bioactive compounds that play a crucial role as antiparasitic agents. Tannins one of the most important phytochemical forms complexes with proteins reducing the availability of nutrients to the worms. They can also effect the worm's metabolism and reproductive capabilities (Nagaradoum, 2018). For instance, the tannin containing ethanoic and aqueous extracts of *Artemisia vulgaris* have also been evaluated by Karim *et al.* (2019) and these demonstrated a significant activity against the eggs, larvae (L3) and adult worms. The ethanolic extracts completely inhibit egg hatching at a concentration of 5 mg/ml while aqueous extracts required a higher concentration of 10 mg/ml to achieve the similar effects. The extracts of *Artemisia herba alba* and *Punctia granatum* demonstrated promising anthelmintic activity at all dose levels in a concentration and time dependent fashion (Hassan *et al.*, 2020). Mravčáková *et al.* (2021) surveyed that the treatment with the *Artemisia absinthium* and *Malva sylvestris* affects antioxidant parameters by reducing oxidative stress in the abomasal mucosa and helps to trigger the local immune responses.

Alkaloids have been reported to interfere with nervous system of parasites resulting in the paralysis and subsequent mortality. Notably alkaloids derived from the *Derris elliptica* and *Sophora flavescens* have exhibited significant anthelmintic properties. Similarly, flavonoids like quercetin which are abundantly distributed in medicinal plants

influence parasitic metabolism, reproductive functions and antioxidant defense mechanisms (Ali *et al.*, 2021). Saponins have also been reported in medicinal plants and these disrupt the integrity of the nematode cell membrane leading to the cell lysis. Some legumes such as fenugreek (*Trigonella foenum graecum*) and Spindaceae family contain saponins with anthelmintic activity (Páez-Léon *et al.*, 2022).

Artemisia vestita has been evaluated by Dogra *et al.* (2023) and a total of 202 bioactive compounds belonging to different chemical classes were reported which are responsible for different pharmacological activities. Another one of the important components found in the medicinal plants are essential oils which are volatile compounds and can disrupt the worm's cellular layer and interfere with various physiological processes. The essential oils of leaves and flowers of *Lipia sidoides* have been tested in vitro using *H. contortus* as the experimental worm and the leaves give a remarkable inhibitory impact on worm hatching in comparison to essential oils from flowers (Ramadani *et al.*, 2023). The presence of fatty acids and terpenes evaluated through the phytochemical analysis of *Artemisia cina* extract has been associated with the higher ovicidal activity of the extract even at the low concentrations (Arango De la Pava *et al.*, 2024). Synergistic interaction effect of *Artemisia cina* and *Tagetes lucida* ethyl extract has been examined by Figueroa *et al.* (2024) and the results demonstrated high egg hatching inhibition with 50/50 combination and the best larvae mortality with 25% *A. cina* and 75% *T. lucida* at 10 mg/ml. In a recent study on oxidative stress on *H. contortus* larvae exposed to *A. cina* n-hexane extract carried out by Sánchez-Mendoza *et al.* (2025) and it has been deduced that there is upregulation of glutathione peroxidase (GPX) and catalase (CAT) and decreased expression of superoxide dismutase (SOD) genes after exposure of *H. contortus* (L3) to n-hexane extract of *A. cina*.

Previous studies have demonstrated that various species of *Artemisia* including *Artemisia vestita* possess bioactive compounds with ovicidal, larvicidal and adulticidal effects against *H. contortus* in sheep and goats. However further investigations are required regarding the efficacy of *A. vestita* under field conditions. There is a gap in *in vivo* trials especially with *A. vestita* and in understanding how extract type, dose, administration and safety affect the outcomes. Another major gap is the lack of systematic evaluation of solvent-based differences; although methanolic extracts are often reported to be more potent due to their ability to solubilize bioactive compounds, comprehensive

comparisons with aqueous extracts are scarce. The present study provides the exclusive phytochemical characterization of *A. vestita* extracts and its *invitro* effectiveness against *H. contortus* in sheep under controlled conditions to validate its potential as an alternative to the synthetic anthelmintic drugs. The main focus of study is to check the efficacy of aqueous and methanolic extracts of *A. vestita* on the adult worm mortality.

3. MATERIALS AND METHODS

3.1 MATERIALS

3.1.1 EXPERIMENTAL ANIMAL

H. contortus is a blood sucking intestinal helminth that lives in the abomasum of small ruminants (Irfan-ur-Rauf Tak *et al.*, 2014). It is commonly known as the twisted barber pole worm or stomach worm and is a causative agent of haemonchosis (Saminathan *et al.*, 2015).

Taxonomic classification

Kingdom: Animalia

Phylum: Nematoda

Class: Secernentea

Family: Strongylida

Genus: *Haemonchus*

Species: *contortus*

Morphology

The body is cylindrical and the blood this species sucks gives the transparent worm a striking reddish color. The female is longer (18-30 mm) than the males (10-20 mm) and their red and white appearance is due to their white ovaries twisted around the blood filled intestines. The buccal capsule is well developed and has cutting plates and lancets that enable the worm to puncture blood vessels and feed on blood (Scott and Sutherland, 2009). A pair of wedge-shaped cervical papillae are present near the anterior end and anal opening is positioned towards the posterior side. The bursa in males is well developed consisting of two lateral lobes along with a single asymmetrically located dorsal lobe which is supported by a Y- shaped dorsal ray (Fig. 2). The vulva of the female is usually covered by a linguiform process (vulvar flap) which is usually large and very prominent (Fig. 3).



Fig. 2. Microscopic images of *Haemonchus contortus* (A) Anterior end of worm (B) Anterior end showing buccal capsule and cervical papillae (40X) (C) Male bursa with two lateral lobes and Y-shaped dorsal ray

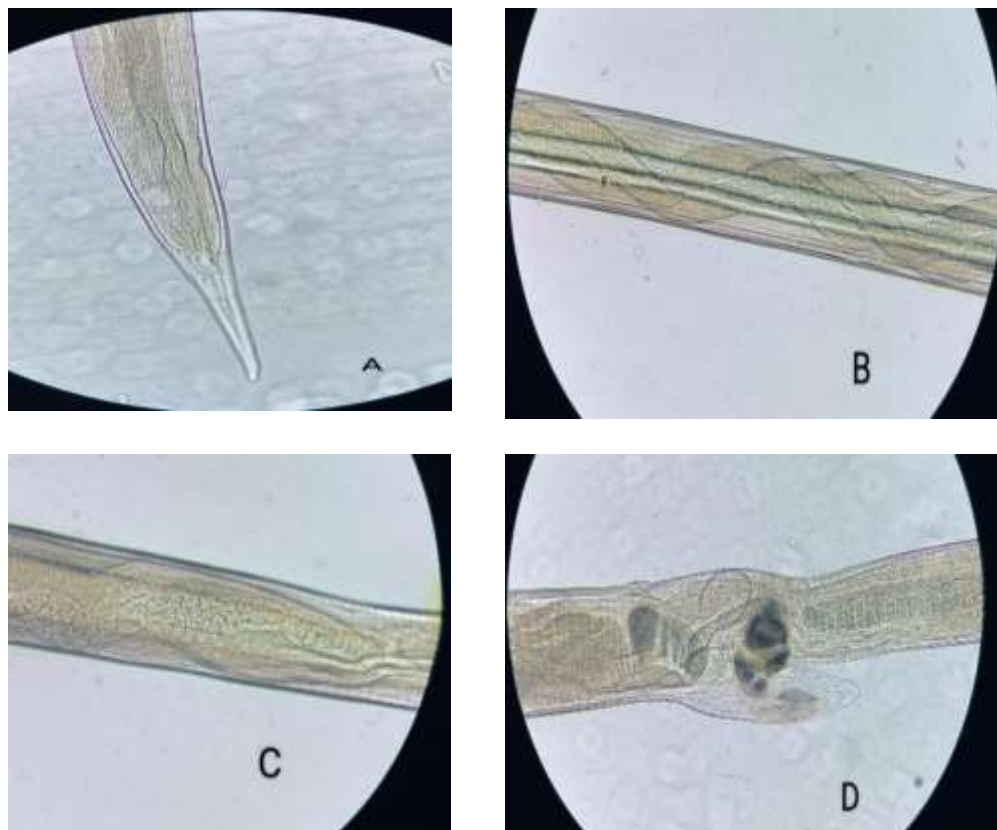


Fig. 3. (A) Posterior end of female (B) Intertwined ovaries and intestines of female (C) Eggs in uterus (D) Linguiform vulva

3.1.2 EXPERIMENTAL PLANT

Artemisia vestita commonly known as Russian wormwood is a herbaceous perennial flowering plant in the family Asteraceae. It is popularly known as Kubsha (Dogra *et al.*, 2023) and is distributed in East Asia including the Himalayas (India, Nepal, Tibet, Pakistan and China). It is a traditional medicinal herb and in Kashmir it grows on exposed hilly slopes at lower altitudes and has been collected from areas like Sonamarg, Zabarwan and Mansbal.

Taxonomic classification

Kingdom: Plantae

Phylum: Streptophyta

Class: Equisetopsida

Order: Asterales

Family: Asteraceae

Genus: *Artemisia*

Species: *vestita*

Morphology

A. vestita is an erect, perennial, aromatic shrub, reaching up to 2 m tall with fern like pinnately cut leaves that soft and hairy. The stem is also covered with fine hairs giving the plant a woolly texture and silver sheen. The stem is strongly aromatic and can be source of essential oils. Its small creamy yellow flowers are arranged in long hairy hanging racemes. The fruits are shiny and smooth with membranes oblong bracts. The plant is hermaphroditic, and the seed ripening takes place from August to September.

3.2 METHODS

3.2.1 COLLECTION OF PLANT MATERIAL

Fresh parts of *Artemisia vestita* were collected from Safapora, Ganderbal (Lat 34.255381° Long 74.689758°) (Fig. 4). The identity of the plant was ascertained based on traditional ethnobotanical knowledge provided by residents. After collection the plant material was thoroughly washed with distilled water to remove adhering soil and debris. After washing the plant material was shade dried for 5 days and then ground into a fine powder using mortar and pestle (Fig. 5). The powdered material was weighed and stored in airtight containers at room temperature until subsequent extraction.



Fig .4. GPS location of plant sampling site



Fig. 5 (A). Dried plant material (B) Grinding plant material using mortar and pestle (C) Ground plant material (D) Weighing of sample before extraction

3.2.2 PREPARATION OF PLANT EXTRACT

The plant extract was prepared using the maceration technique as previously reported by (Metoui *et al.*, 2022) with slight modifications. Two solvents were used – distilled water (aqueous) and methanol (alcoholic). The powdered *A. vestita* dissolved in both solvents separately in the ratio of 1:5 (W/V). The plant material was completely immersed in the solvent, and the container was sealed to prevent evaporation. The mixture was gradually heated at 55°C for about four to six hours (Fig.6). To enhance extraction the mixture was occasionally shaken. After maceration the solvent was filtered through the muslin cloth and then again subjected to filtration through Whatman no.1 filter paper (Fig.7). Some of the crude liquid extract was stored for the subsequent phytochemical analysis and the rest of the filtrate was concentrated by evaporating the solvent in a water bath at controlled temperature (Fig. 8). (All the above mentioned steps were performed separately for both the solvents).

The semi-solid residues obtained after solvent evaporation were weighed and the yield of the aqueous extract was 5g while that of methanolic extract was 3g. Accordingly various concentrations of both aqueous and methanolic extracts of *A. vestita* were prepared for experimental assays. The aqueous extract was dissolved in PBS (Phosphate buffered saline pH~7.4) to obtain the desired concentrations (100, 60, 30, 15 mg/ml). In case of methanolic extract the desired concentrations (100, 60, 30, 15 mg/ml) were prepared using PBS with the addition of 5% DMSO (Dimethyl sulfoxide) as a co-solvent to ensure solubility. Also 5 mg/ml solution of albendazole was prepared as a positive control (Fig. 9).



Fig.6. (A) Heating plant material in methanol (B) Heating plant material in water



Fig. 7. (A) and (B) Filtration of solvent extract of plant material

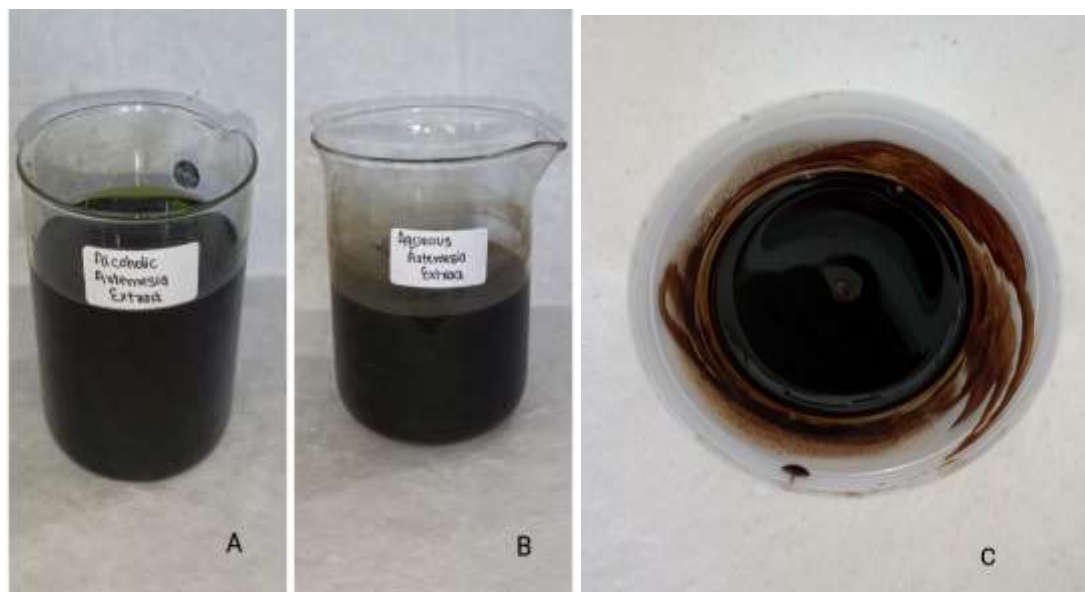


Fig. 8. (A) Crude methanolic (alcoholic) filtrate (B) Crude aqueous filtrate (C) Semisolid extract obtained after concentration

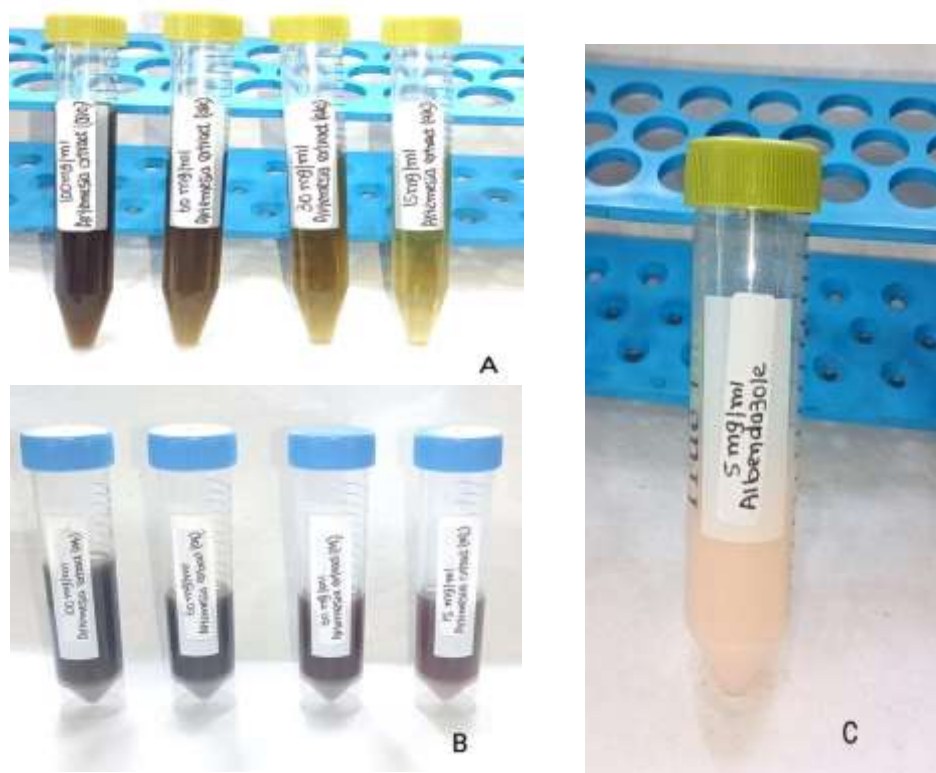


Fig. 9. (A) Working concentrations of methanol extract (B) Working concentrations of aqueous extract (C) Albendazole as positive control

3.2.3 PHYTOCHEMICAL ANALYSIS OF *Artemisia vestita*

Preliminary phytochemical screening of the aqueous and methanolic extracts of *Artemisia vestita* was conducted to determine the presence of major bio active compounds (Fig.10). The crude filtrate obtained from both aqueous and methanolic extracts was subjected to 9 preliminary qualitative assays to determine the presence of various phytochemical constituents. These assays were performed following the standard protocols described by Sheikh and Patil (2020). The assays performed and their corresponding targeted phytochemicals are summarized in Table1. The phytochemical screening results of *A. vestita* are presented in Table 2.

The phytochemical screening revealed the presence of multiple classes of secondary metabolites in both aqueous and methanolic extracts. The assays indicated that alkaloids, flavonoids, tannins and phenolic compounds were present in both extracts as confirmed by the characteristic precipitate formation or colour changes (Fig.11). Terpinoids and quinones were strongly expressed in the methanolic extract as compared to aqueous extract.

Table 1: Qualitative tests for phytochemical screening

Test	Procedure	Observations (indicating positive results)
	Detection of alkaloids	
Picric acid test	Few ml filtrate + 1-2 drops of 2% picric acid solution	An orange colour
	Detection of glycosides	
Keller-Killani test	1ml filtrate + 1.5 ml glacial acetic acid + 1 drop of ferric chloride + concentrated H ₂ SO ₄ (along the side of test tube)	A blue coloured solution (in acetic acid layer)
Baljet test	2ml extract + a drop of Baljet's reagent (19ml of 1% picric acid + 1ml 10% NaOH)	A yellow orange colour
	Detection of flavonoids	
Alkaline reagent test	1ml extract + 2ml of 2% NaOH (+ few drops of dilute HCl)	An intense yellow colour , becomes colourless on addition of diluted HCl
Ferric chloride test	few ml extract + few drops of 10% ferric chloride solution	A green precipitate
	Detection of phenolic compounds	
Gelatin test	Few ml of plant extract +1ml of 1% gelatin solution + few drops of 10% NaCl	A white precipitate
	Detection of tannins	
Braymer's test	1ml extract 3ml distilled water + 3 drops of 10% ferric chloride solution	Blue green colour
	Detection of terpinoids	
Salkowski's test	1ml filtrate + few drops of conc. H ₂ SO ₄ (shaken well and allowed to stand)	Golden yellow layer at the bottom
	Detection of quinones	
Conc. HCl test	1ml extract + a drop of conc. HCl	A green colour

Table 2: Results showing presence (+) and absence (-) of various phytochemicals in methanol and aqueous extracts of *A. vestita*

Sno.	Test	Methanolic extract	Aqueous extract
1	Picric acid test	+	+
2	Keller-Killani test	—	+
3	Baljet test	+	+
4	Alkaline reagent test	+	—
5	Ferric chloride test	+	+
6	Gelatin test	+	+
7	Braymer's test	+	+
8	Salkowski's test	+	+
9	Conc. HCl test	+	—



Fig. 10. (A) and (B) Working solutions of chemicals for phytochemical analysis (C) Performing phytochemical tests



Fig.11. Test tubes showing phytochemical reactions of (A) Aqueous *A. vestita* extract and (B) Methanolic *A. vestita* extract with different chemicals

3.2.4 ISOLATION OF *Haemonchus contortus* FROM SHEEP ABOMASUM

Abomasal samples of sheep were procured from a local butcher immediately after slaughter and transported to the laboratory in clear container (Fig.12). The abomasa were carefully dissected along the greater curvature using a sterile scalpel and the contents were emptied into the beaker. The mucosal lining of abomasum was thoroughly examined for the *Haemonchus contortus*. Adult live specimens of *H. contortus* were recovered and collected in PBS (pH~7.4) (Fig.12), then subsequently identified on the basis of morphological characteristics including the spiral red and white barber's pole appearance of females, the presence of a well developed copulating bursa in males and prominent cervical papillae. The diagnostic criteria followed were in accordance with the description provided by Amarante *et al.* (2011).



Fig. 12. (A) Abomasum of sheep (B) Dissected abomasum (C) Recovering *Haemonchus contortus* (D) Collected worms in PBS (pH~7.4)

3.3 ADULT MORTALITY ASSAY

The *invitro* adult mortality assay was conducted to evaluate the anthelmintic potential of the test extracts against adult *H. contortus*. Mature live worms were used to evaluate the effect of aqueous and methanolic extracts of *A. vestita* following the protocol described by Mumed *et al.* (2022) with minor modifications. The worms were placed in 24 microwell plate containing different concentrations of aqueous and methanolic extracts prepared in PBS and PBS + DMSO respectively (Fig.13). Each concentration was tested in triplicate with PBS and DMSO serving as negative controls while as the albendazole (5mg/ml) was employed as the positive control. The 24 microwell plate was then placed in a BOD incubator and maintained at 37°C to simulate the physiological conditions of the host. The worms were monitored at regular intervals (0, 1, 2, 3, 4, 5, 6 hours) to assess motility and mortality. The number of worms showing no movement even upon gentle probing were considered dead and were recorded for each treatment.

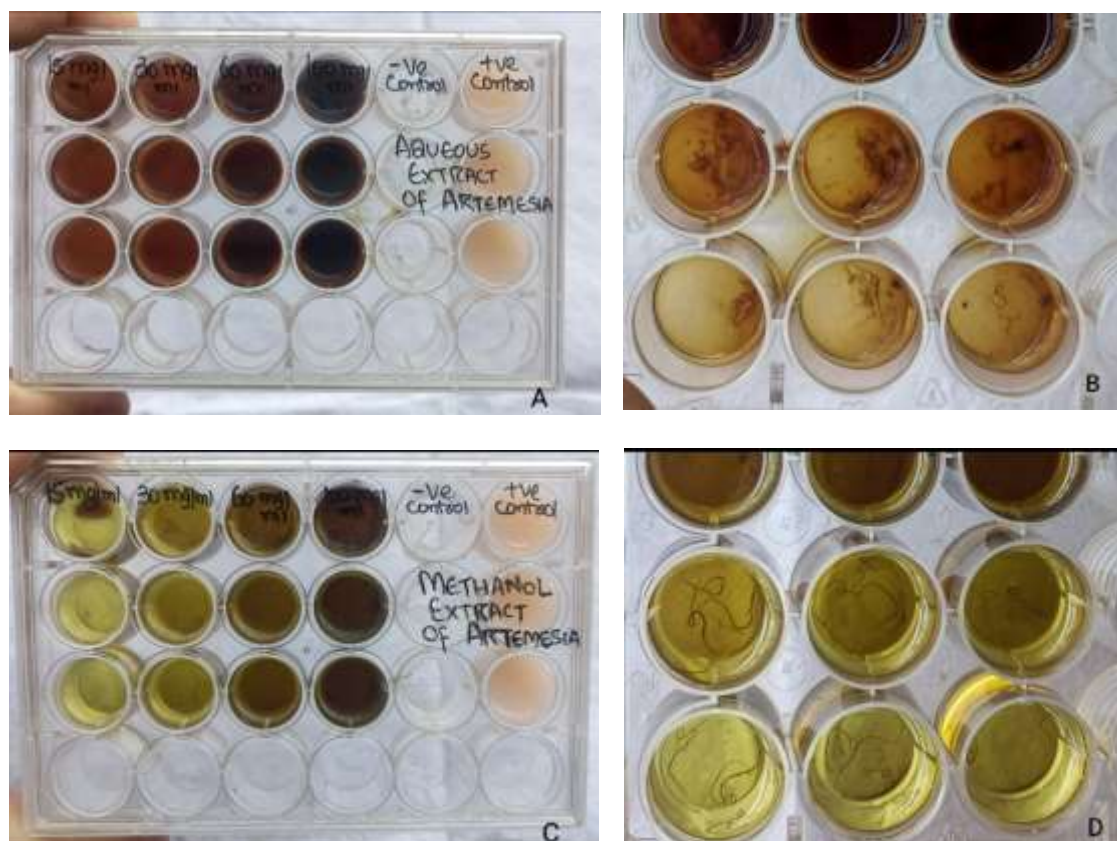


Fig. 13. (A) Treatment of worms with different concentrations of aqueous extract of *A. vestita* (B) Dead and disintegrated worms in aqueous extract (C) Treatment of worms with different concentrations of methanolic extract of *A. vestita* (D) Dead and disintegrated worms in methanolic extract

4. RESULTS

The adult mortality assay demonstrated a concentration and time dependent effect of both the extracts on *H. contortus*. Mortality increased with both extract concentration and incubation time. The methanol extract at highest concentration (100 mg/ ml) induced 100% mortality within 4 hours. At the lower concentrations mortality was 90-95% at 60 mg/ml, 70-73% at 30 mg/ml and approximately 40-45% at 15 mg/ml. In contrast the aqueous extract at its highest concentration (100 mg/ml) achieved 100% mortality after 6 hours and at the lower concentrations the mortality was 60-65% at 60 mg/ml, 50-53% at 30 mg/ml and 30-35% at 15 mg/ml. As expected, the highest mortality was observed in the positive control i.e., albendazole (5 mg/ml) which induced complete death of all worms within 2-3 hours. In the negative controls PBS and DMSO a small proportion of worms (10-15%) died after 6 hours likely due to handling stress and prolonged incubation. All results are presented in the form of tables (Table 3 and 4) and graphs (Fig.14 and Fig.15) illustrating the differences in mortality between the various treatments and controls.

Table 3: Mean number of worms showing mortality post exposure to various concentrations of methanolic extract of *A. vestita*

Methanol extract (mg/ml)	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD
	0hr	1hr	2hr	3hr	4hr	5hr	6hr
100	0 $\pm$ 0.00	4.00 $\pm$ 1.00	4.66 $\pm$ 0.57	5.00 $\pm$ 0.00	5.00 $\pm$ 0.00	5.00 $\pm$ 0.00	5.00 $\pm$ 0.00
60	0 $\pm$ 0.00	2.33 $\pm$ 0.57	3.00 $\pm$ 0.57	3.66 $\pm$ 0.57	4.33 $\pm$ 0.57	5.00 $\pm$ 0.00	5.00 $\pm$ 0.00
30	0 $\pm$ 0.00	1.66 $\pm$ 0.57	1.66 $\pm$ 0.57	2.00 $\pm$ 0.00	2.66 $\pm$ 0.57	3.00 $\pm$ 1.00	3.66 $\pm$ 0.57
15	0 $\pm$ 0.00	0.66 $\pm$ 0.57	1.33 $\pm$ 0.57	1.33 $\pm$ 0.57	1.66 $\pm$ 0.57	2.33 $\pm$ 0.57	2.66 $\pm$ 0.57
Positive control	0 $\pm$ 0.00	4.00 $\pm$ 1.00	4.33 $\pm$ 0.57	5.00 $\pm$ 0.00	5.00 $\pm$ 0.00	5.00 $\pm$ 0.00	5.00 $\pm$ 0.00
Negative control	0 $\pm$ 0.00	0.33 $\pm$ 0.57	0.66 $\pm$ 0.57	0.66 $\pm$ 0.57	1.00 $\pm$ 0.00	1.33 $\pm$ 0.57	1.66 $\pm$ 0.57

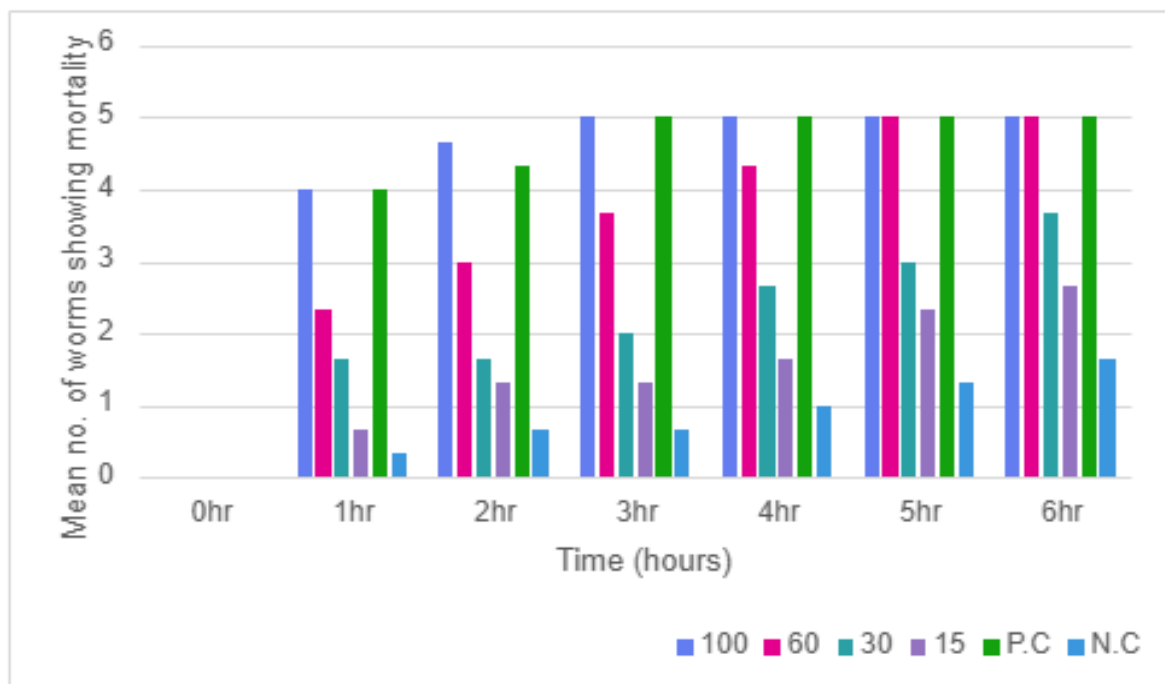


Fig. 14. Graphical representation of table 3

Table 4: Mean number of worms showing mortality post exposure to various concentrations of aqueous extract of *A. vestita*

Aqueous extract (mg/ml)	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD
	0hr	1hr	2hr	3hr	4hr	5hr	6hr
100	0 $\pm$ 0.00	3.66 $\pm$ 0.57	4.00 $\pm$ 1.00	4.33 $\pm$ 0.57	4.66 $\pm$ 0.57	5.00 $\pm$ 0.00	5.00 $\pm$ 0.00
60	0 $\pm$ 0.00	2.00 $\pm$ 0.00	2.33 $\pm$ 0.57	2.33 $\pm$ 0.57	2.66 $\pm$ 0.57	3.33 $\pm$ 0.57	4.00 $\pm$ 1.00
30	0 $\pm$ 0.00	1.33 $\pm$ 0.57	1.66 $\pm$ 0.57	1.66 $\pm$ 0.57	2.00 $\pm$ 0.00	2.66 $\pm$ 0.57	3.33 $\pm$ 0.57
15	0 $\pm$ 0.00	0.33 $\pm$ 0.57	0.66 $\pm$ 0.57	1.33 $\pm$ 0.57	1.66 $\pm$ 0.57	1.66 $\pm$ 0.57	2.00 $\pm$ 0.00
Positive control	0 $\pm$ 0.00	4.00 $\pm$ 1.00	4.33 $\pm$ 0.57	5.00 $\pm$ 0.00	5.00 $\pm$ 0.00	5.00 $\pm$ 0.00	5.00 $\pm$ 0.00
Negative control	0 $\pm$ 0.00	0.33 $\pm$ 0.57	0.33 $\pm$ 0.57	0.66 $\pm$ 0.57	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.33 $\pm$ 0.57

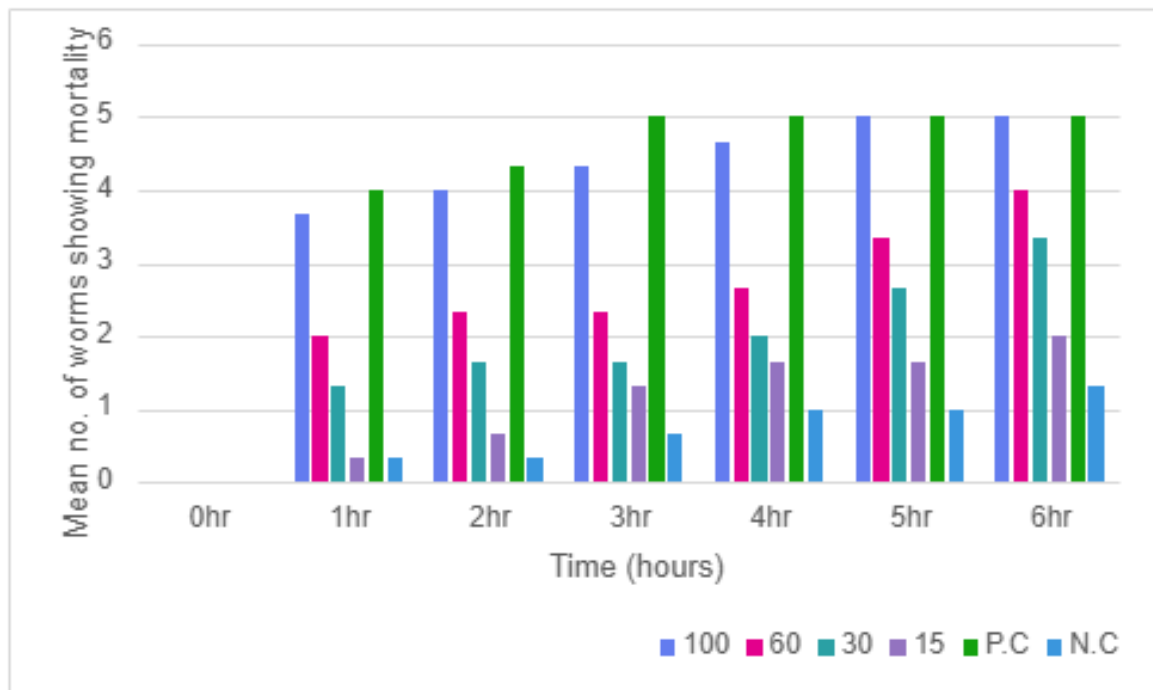


Fig. 15. Graphical representation of table 4

5. DISCUSSION

The aqueous and the methanolic extracts derived from *A. vestita* exhibited considerable effectiveness against the *H. contortus*. The findings suggest that *A. vestita* contains phytochemical compounds with significant anthelmintic potential capable of suppressing the parasite successfully. The efficacy observed is likely attributed to the solvents - distilled water and methanol which enabled the extraction and retention of these phytochemicals. Such outcomes highlight the potential of *A. vestita* as natural and alternative approach for the control of *H. contortus* infections, offering a new perspective in the development of plant based anthelmintic agents. The results indicate that the methanolic extract is more potent than the aqueous extract producing faster and higher mortality at comparable concentrations. The higher efficacy of the methanolic extracts may be attributed to its ability to extract a wide range of secondary metabolites like flavonoids and tannins which are absent or less concentrated in aqueous extract. Similar findings were reported by many researchers exploring the anthelmintic properties of various medicinal plants.

The results of the current study align with the observations reported by De Jesus-Martinez *et al.* (2018) who demonstrated that the methanolic extract of *Caesalpinia coriaria* produces strong lethal effects on *H. contortus*. Khan *et al.* (2024) also reported that the methanolic extract from *Vitex negundo* leaves is more effective against *H. contortus* than the pure aqueous extracts. Notably studies on the plants belonging to the genus *Artemisia* have revealed their efficacy in combating gastrointestinal parasites. The research carried out by Tariq *et al.* (2009) demonstrated that the crude alcoholic (ethanol) extract of *A. absinthium* is more potent than the crude aqueous extract against the ovine nematodes. Akkari *et al.* (2014) reported that the ethanolic extract of *A. campestris* exhibited greater efficacy against *H. contortus* than that of the aqueous extract. A study by Irum *et al.* (2015) demonstrated that the methanolic extracts of *A. vestita* and *A. maritima* induced significant larval and adult mortality in *H. contortus*. The above mentioned references highlight the growing interest in utilizing plant derived compounds as sustainable alternatives. *A. vestita* stands out as a plant with promising bioactive components that may serve as a foundation for developing the new anthelmintic treatment.

6. CONCLUSION

The primary focus of the current study is on evaluating the phytochemical composition of *A. vestita* and its anthelmintic effects on *H. contortus* - a blood sucking nematode known to infect the abomasum of small ruminants. The results of the phytochemical assays demonstrated the presence of many bioactive compounds which are responsible for the observed anthelmintic activity. The aim of this research is to elucidate and provide evidence for the anthelmintic potential of experimental plant in controlling the infections caused by the targeted nematode specie. The findings of the adult mortality assay revealed that the methanolic extract exhibited superior efficacy compared to the aqueous extract. The findings of the phytochemical assay and adult mortality assay suggest a strong correlation between the presence of the phytochemical compounds and the enhanced efficacy of the methanolic extract.

Furthermore, this study is anticipated to contribute insights into the practical application of *A. vestita* as a potential natural solution for managing parasitic infections. While the *invitro* results are promising further *invivo* studies are recommended to validate the potential and safety of the extracts for the host animal. More research to identify the specific compounds responsible for the observed effects as well as to understand their mechanism of action will be essential for advancing our knowledge of plant based treatments of parasitic infections.

In conclusion the study establishes *A. vestita* as a promising source of phytochemical compounds reinforcing the potential of plant based alternatives in shaping natural sustainable approaches for the management of helminth infections in animals. However, exploration of formulation and dosage optimization is recommended for the potential therapeutic application in veterinary practice.

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